

Working Title: A Study of Low-Light Enhancement in the HVI Color Space

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Abstract

HVI-CIDNet (Yan et al., 2025) enhances low-light images by mapping them into the HVI (Horizontal/Vertical/Intensity) color space, processing intensity and chroma along separate branches, and supervising the output in both RGB and HVI. This thesis studies a question about that design; the question, the claims, and the experiments are fixed in the project plan before any run starts, and every reported number is read from a result file by a verification script rather than transcribed.

1. Introduction

Low-light image enhancement recovers a well-exposed image from an underexposed capture. Learned enhancers now dominate the paired benchmarks (Wei et al., 2018; Cai et al., 2023; Guo et al., 2020), and a recent line of work argues that the color space in which the network operates and is supervised matters as much as the architecture. HVI-CIDNet (Yan et al., 2025) maps images into a polar Horizontal/Vertical/Intensity (HVI) space, enhances intensity and chroma along separate interacting branches, and is trained with a loss summed over both the RGB output and its HVI transform.

This paper asks one question about that design. The question, the claims it supports, and the experiments that decide it are stated in Section 2; the results follow in Section 3.

2. Method and Protocol

Model. We use the public HVI-CIDNet implementation (Yan et al., 2025). The network takes an sRGB image, converts it to HVI, processes the intensity map and the chroma map along two interacting branches, and converts the result back to sRGB.

Protocol. All runs share one frozen protocol: the same dataset split, seeds, optimizer, schedule, and hardware (a

single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32). Checkpoints are selected on the validation split only; the test split is read once, by a separate final-evaluation script, per reported configuration. Each run records the SHA-256 of its training script, the upstream commit, and the split file.

3. Results

Results are reported here once the studies defined in the project plan have run. Every table in this section is generated from a result file, and every number is bound to the key it was read from.

4. Conclusion

We study one question about the HVI-CIDNet design under a single frozen protocol, and report every finding against its seed spread. The protocol and scripts are released with the thesis.

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A. Related Work

Learned low-light enhancement. Retinex-inspired decomposition networks (Wei et al., 2018), curve-estimation methods that need no paired data (Guo et al., 2020), and transformer enhancers (Cai et al., 2023) form the main families evaluated on paired low-light benchmarks. HVI-CIDNet (Yan et al., 2025) differs by changing the color space in which enhancement and supervision happen rather than the backbone.

Perceptual and structural losses. SSIM (Wang et al., 2004), VGG feature losses (Johnson et al., 2016), and LPIPS (Zhang et al., 2018) are standard components and evaluation metrics for restoration.

B. Additional Experiments

Full tables, per-image analysis, and protocol details are reported here once the studies have run.

B.1. Protocol Details

Each run directory records the training-script SHA-256, the upstream HVI-CIDNet commit, the split-file SHA-256, the torch version, and the GPU name. The validation summary written at the end of every run is the only source of the numbers in this paper.